Sovereign by Design: An Integrated Five-Pillar Framework for India's Strategic Compute and Artificial Intelligence Infrastructure

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***Abstract*—National AI sovereignty requires co-optimizing physical, microelectronic, and institutional infrastructure. Prevailing policy treats compute, power grids, data centres, connectivity, and governance in isolation, creating capital misallocations and grid vulnerabilities. This paper introduces the India AI Infrastructure Readiness Framework (IAIRF 2.0) and an auditable six-layer Demand-Cascade Model (DCM) parameterised using verified 2026 regulatory data. Four key findings emerge. First, deploying 500,000 accelerators demands 891.8 MW of peak facility power, absorbing over 56% of India's operational colocation capacity. Second, domestic sub-5 nm wafer fabrication is not the binding constraint: required logic wafer starts consume only 1.60% of global foundry capacity, while CoWoS advanced packaging consumes 7.41%—a 4.6× higher constraint intensity. Third, direct-to-chip liquid cooling slashes evaporative water withdrawal by 94.6% and reduces facility power by 172 MW. Fourth, Mumbai's 52.4% capacity concentration and 20–40% dynamic AI load swings threaten transmission stability. A sequenced three-tier roadmap prioritising transmission corridors, packaging partnerships, and regional dispersion is presented.**

***Index Terms*—Advanced packaging, Artificial intelligence, Compute sovereignty, Data centre cooling, High-bandwidth memory, Power grid infrastructure.**

I. Introduction

AI has transformed from a research discipline into an industrial apparatus determined primarily by capital, high-voltage grid infrastructure, and geopolitically concentrated microelectronics [1]. Global data-centre electricity consumption reached 415 TWh in 2024; the IEA projects it will surpass 945 TWh by 2030 [2]. Data-centre power grows at 30% annually—four times the rate of new grid generation capacity [3].

India holds indisputable strategic assets: 4.3 million software engineers, the world's highest AI-skill penetration rate [4], and a geopolitical position coveted by both Washington and Beijing [5]. The IndiaAI Mission (Rs. 10,371 crore, approved March 2024) [6] and Semicon India 2.0 (Rs. 1,27,500 crore, approved July 2026) [7] codify national ambition. By mid-2026, the shared GPU pool reached 45,000 units at Rs. 65/GPU-hour, supporting 237 research projects. India's operational colocation capacity reached nearly 1.8 GW across 271 facilities by H1 2026 [8], [9].

The critical analytical deficit is *disciplinary siloing*: semiconductor analysts study supply chains without modelling data-centre power; grid researchers model TWh demand without tracing packaging constraints; governance scholars enumerate policy pillars without physical resource accounting. This paper closes that gap with two original contributions:

- **C1. IAIRF 2.0**: A Five-Pillar framework mapping causal interdependencies across compute, energy, physical resources, connectivity, and governance.
- **C2. DCM**: A reproducible six-layer Demand-Cascade Model propagating GPU targets through server physics, facility thermodynamics, cooling hydrology, wafer geometry, advanced packaging, and HBM allocation.

II. Literature Review and Research Gaps

A. Compute and Semiconductor Supply Chains

Akimitsu [1] characterises the AI hardware supply chain as a directed acyclic dependency graph with extreme upstream concentration: EUV lithography (ASML monopoly, HHI = 10,000), CoWoS packaging (TSMC ≈90%,

TABLE I
PRIOR ART COMPARISON MATRIX

Source	Method	GPU Scen.	Gaps Closed
Akimitsu (2026) [1]	HHI graph	None	RG-03 partial
IEA/LBNL (2024) [2], [19]	Energy model	Global	RG-04 partial
Kaufmann (2025) [20]	Grid power-flow	None	RG-01 partial
SBI CHAKRA (2025) [16]	Market census	None	RG-04 partial
This paper (IAIRF 2.0)	**5-Pillar+DCM**	**200k–1M**	**RG-01–05**

HHI = 8,100), refined gallium (China ≈99%), and HBM (SK Hynix ≈57%, Samsung ≈38%, Micron ≈5%) [10]. CSET [11] establishes that >90% of sub-5 nm logic fabrication resides in Taiwan. Under Semicon India 2.0, India has approved 12 semiconductor projects including Tata Electronics' ATMP at Morigaon and Tata-PSMC's 28/50/90 nm fab at Dholera [12].

B. Energy and Grid Readiness

AI training racks draw 60–100 kW/rack (H100) to 120–200 kW/rack (Blackwell) [13]. Grid-connection queues for >50 MW loads extend 18–36 months [14]. The IEGC mandates a tight 49.90–50.05 Hz operating band; AI workloads exhibit 20–40% sub-minute power swings. Under the swing equation:

$$\frac{2H}{f_0}\frac{df}{dt} = P_m - P_e - \Delta P_{\text{AI}} \tag{1}$$

a 250 MW step-load shed in a 50 GW grid ($H = 4.2$ s) induces RoCoF of -0.0298 Hz/s, risking UFR tripping without fast BESS response [15].

C. Data Centres, Connectivity, and Research Gaps

India's colocation capacity grew from 375 MW (2020) to nearly 1.8 GW (H1 2026) across 271 facilities, with 52.4% concentrated in Mumbai [8]. Indian data centres consumed ≈150 billion litres of water in 2024, with average PUE of 1.55 under tropical conditions [9], [16]. Indic script tokenization imposes a 3.3×–5.6× token expansion over Latin text; since self-attention complexity scales as $\mathcal{O}(L^2)$, this creates a 10.9×–20.3× compute penalty [17]. The CEA Cyber Security Regulations 2026 mandate 6-hour incident reporting and domestic data residency for ≥50 MW assets [18]. Five gaps remain (Table I): disciplinary siloing (RG-01), absent demand cascade (RG-02), silicon wafer fallacy (RG-03), tropical cooling underestimation (RG-04), and operationalisation deficit (RG-05).

III. THE INDIA AI INFRASTRUCTURE READINESS FRAMEWORK 2.0

IAIRF 2.0 organises India's challenge across five causally interdependent pillars:

- **P1 — Compute & Semiconductor Sovereignty:** GPU procurement, ATMP/OSAT packaging, critical mineral supply chains. Sovereign pool: 45,000+ accelerators at Rs. 65/GPU-hr [21].
- **P2 — Energy & Grid Readiness:** Data-centre electrical load, grid interconnection lead times, IEGC frequency stability, 20–40% AI load swings [15].
- **P3 — Physical Resources & Data Centres:** Facility capacity, cooling thermodynamics, PUE, water withdrawal, geographic concentration [8], [9].
- **P4 — Connectivity, Data & Security:** Submarine cable infrastructure (17+ systems), DPDPA compliance, OT cybersecurity under CEA 2026 regulations [18], [22].
- **P5 — Governance, Investment & Indigenous Capability:** IndiaAI Mission, Semicon India 2.0, BharatGen foundation models, Pax Silica/iCET alliances [7], [23].

Four systemic feedback loops constrain the framework: **Loop A** (Power-Compute Deadlock: GPU procurement blocked by facility power, grid lead times, and land acquisition); **Loop B** (Packaging-Memory Co-Constraint: EAR changes sever CoWoS and HBM3e dependencies simultaneously); **Loop C** (Data-Connectivity Governance Tension: DPDPA localisation raises infrastructure costs and foreign investment uncertainty); **Loop D** (Talent-Sovereignty Paradox: 4.3M engineers disproportionately serve foreign GCCs, with AI/ML talent gap of ≈720,000 widening to 1.6M by 2030 [4]).

IV. DEMAND-CASCADE MODEL FORMULATION

The DCM maps an accelerator target N (GPUs) through six deterministic computational layers grounded in verified empirical parameters (Fig. 1).

A. Layer 1: IT Electrical Load

Active IT power uses reference TDP $P_{\text{gpu}} = 700$ W (NVIDIA H100 SXM5 [13]) and server chassis overhead $R_{\text{host}} = 1.82$:

$$P_{\text{IT}} = \frac{N \cdot P_{\text{gpu}} \cdot R_{\text{host}}}{10^6} \quad [\text{MW}] \tag{2}$$

B. Layer 2: Facility Power and Annual Energy

Facility draw uses tropical baseline PUE $= 1.40$; annual energy uses 80% utilisation over 8,760 hr:

$$P_{\text{facility}} = P_{\text{IT}} \cdot \text{PUE} \quad [\text{MW}] \tag{3}$$

$$E_{\text{annual}} = \frac{P_{\text{facility}} \cdot 8{,}760 \cdot U}{10^6} \quad [\text{TWh}] \tag{4}$$

GHG emissions apply the CEA grid factor $\text{EF}_{\text{grid}} = 0.716$ tCO_2/MWh [14]:

$$\text{GHG} = E_{\text{annual}} \cdot \text{EF}_{\text{grid}} \quad [\text{Mt CO}_2/\text{yr}] \tag{5}$$

C. Layer 3: Cooling Water and Rack Footprint

Annual evaporative water withdrawal uses $I_{\text{water}} = 1.80$ L/kWh for tropical air cooling [2]:

$$W_{\text{annual}} = \frac{E_{\text{annual}} \cdot 10^9 \cdot I_{\text{water}}}{10^6} \quad [\text{ML}] \tag{6}$$

Physical rack count at $D_{\text{rack}} = 60$ kW/rack:

$$N_{\text{rack}} = \left\lceil \frac{P_{\text{IT}} \times 10^3}{D_{\text{rack}}} \right\rceil \tag{7}$$

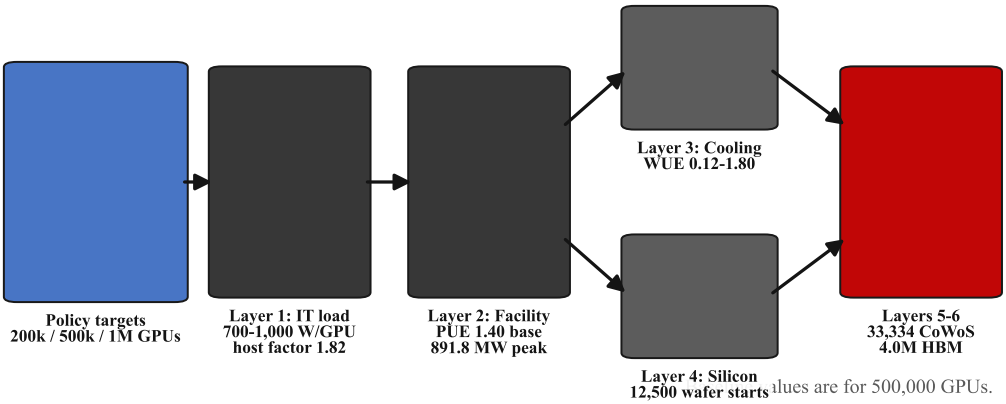


Fig. 1. Six-layer demand-cascade architecture: from national compute policy targets through server electrical load, facility power, cooling thermodynamics, silicon wafer starts, and advanced packaging chokepoints.

D. Layer 4: Semiconductor Wafer Starts

Gross dies per 300 mm wafer ($r = 150$ mm, $A_{\text{die}} = 814$ mm^2, GH100 reticle):

$$N_{\text{gross}} = \left\lfloor \frac{\pi r^2}{A_{\text{die}}} - \frac{2\pi r}{\sqrt{2A_{\text{die}}}} \right\rfloor = 63 \tag{8}$$

Murphy defect-density yield ($D_0 = 0.08$ cm^{-2}, bounds $[0.06, 0.12]$):

$$Y_{\text{Murphy}} = \left(\frac{1 - e^{-D_0 A_{\text{die}}}}{D_0 A_{\text{die}}} \right)^2 \approx 0.540 \tag{9}$$

With industrial binning ($Y = 0.65$, $N_{\text{good}} = 40$ dies/wafer), required wafer starts:

$$W_{\text{starts}} = \left\lceil \frac{N}{N_{\text{good}}} \right\rceil \tag{10}$$

E. Layers 5–6: CoWoS Packaging and HBM

Each GPU requires one CoWoS 2.5D interposer (15 packages/300 mm wafer, $C_{\text{CoWoS}} = 15$):

$$W_{\text{CoWoS}} = \left\lceil \frac{N}{C_{\text{CoWoS}}} \right\rceil \tag{11}$$

Global CoWoS capacity: ≈450,000 wafers/yr; TSMC controls ≈90% (HHI = 8,100) [24]. HBM3e stacks per GPU ($N_{\text{stacks}} = 8$):

$$N_{\text{HBM}} = N \cdot N_{\text{stacks}} \tag{12}$$

Global HBM3e output is ≈60M stacks/yr; India has zero domestic production.

V. Empirical Results and Cross-Pillar Analysis

A. Scenario Simulation Results

Table II presents DCM output across three deployment scenarios and three sensitivity variants. Low variant: $P_{\text{gpu}} = 700$ W, PUE = 1.25; Base variant: $P_{\text{gpu}} = 700$ W, PUE = 1.40; High variant: $P_{\text{gpu}} = 1{,}200$ W, PUE = 1.55. In the baseline 500,000-GPU scenario, active IT load reaches 637.0 MW and peak facility power escalates to **891.8 MW**—equivalent to **56.62%** of India's entire mid-2026 colocation capacity [25]. Annual electricity consumption totals **6.25 TWh/year**, generating 4.47 Mt CO_2, and evaporative water demand reaches **11,249.5 ML/year**. The 1M-GPU high scenario generates 3,255 MW, exceeding India's current capacity by 2.07×. The tornado sensitivity profile in Fig. 3 confirms that facility cooling PUE and server host overhead ratios account for the largest deviations in peak power draw.

B. Finding 1: Refuting the Silicon Wafer Hypothesis

A central premise of sovereign compute policy is that domestic silicon wafer fabrication constitutes the primary bottleneck to sovereign capability. DCM results refute this hypothesis. Deploying 500,000 accelerators requires exactly **12,500** 300 mm wafer starts. Against TSMC's annual sub-5 nm output of ≈780,000 wafers [26], India's requirement represents **1.60%** of global capacity—a volume commercially accessible through capacity reservation contracts. Building a domestic sub-3 nm gigafab requires $15–20 billion in capex, 7–10 years of qualification, and >25,000 wafer-starts/month at minimum efficient scale. Diverting capital to a leading-edge fab to satisfy a 1.60% demand volume represents severe capital misallocation.

C. Finding 2: Structural Chokepoint Inversion

While silicon lithography is abundant relative to demand, Table III and Fig. 2 reveal the true binding constraints are advanced packaging and metropolitan power transmission—not front-end wafer fabrication. **CoWoS Packaging:** 500k GPUs require 33,334 CoWoS interposer wafers, consuming **7.41%** of global capacity (HHI = 8,100)—a constraint intensity **4.6× higher** than front-end wafers. **HBM3e Memory:** Absorbs **6.67%** of annual global output from a triopoly where India has zero domestic production. **Metropolitan Grid Saturation:** Feeding 891.8 MW into Mumbai (52.4% national capacity) requires 18 dedicated 50 MW substations under CEA standards [14]. Developers face a 26-month average wait for 220 kV energization, stranding over 280 MW of newly built IT capacity.

TABLE II
DEMAND-CASCADE MODEL SIMULATION RESULTS ACROSS GPU DEPLOYMENT SCENARIOS (MODEL V2.0) [SOURCES: [19], [24]–[26]]

Scenario	Var.	IT MW	Fac. MW	TWh/yr	Water ML	Wafers	TSMC%	CoWoS	CoWoS%	Colo%
200k GPUs	Low	224.0	280.0	1.59	318.9	4,445	0.57%	11,112	2.47%	17.78%
	Base	254.8	356.7	2.50	4,499.8	5,000	0.64%	13,334	2.96%	22.65%
	High	420.0	651.0	5.13	12,831.9	10,527	1.35%	16,667	3.70%	41.33%
500k GPUs	Low	560.0	700.0	3.99	797.2	11,112	1.42%	27,778	6.17%	44.44%
	Base	637.0	**891.8**	**6.25**	**11,249.5**	12,500	1.60%	33,334	**7.41%**	**56.62%**
	High	1,050.0	1,627.5	12.83	32,079.8	26,316	3.37%	41,667	9.26%	103.33%
1M GPUs	Low	1,120.0	1,400.0	7.97	1,594.3	22,223	2.85%	55,556	12.35%	88.89%
	Base	1,274.0	1,783.6	12.50	22,499.0	25,000	3.21%	66,667	14.81%	113.24%
	High	2,100.0	3,255.0	25.66	64,159.5	52,632	6.75%	83,334	18.52%	206.67%

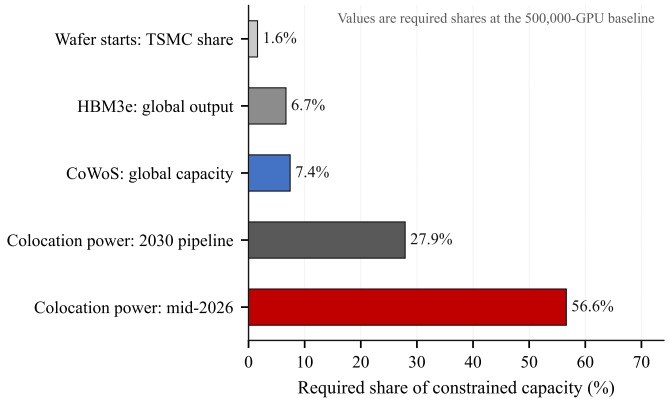


Fig. 2. Structural chokepoint inversion (500,000-GPU baseline): advanced packaging (7.41%) and colocation capacity (56.62%) far outstrip logic wafer requirements (1.60%).

TABLE III
STRUCTURAL CHOKEPOINT HIERARCHY (500K GPU BASELINE)
[SOURCES: [1], [14], [15], [24], [26]]

Infrastructure Layer	HHI	India%	500k%	Lead Time
EUV Lithography (ASML)	10,000	0%	Upstream	24–36 mos
Sub-5nm Logic (TSMC)	5,400	0%	**1.60%**	14–18 wks
CoWoS Packaging (TSMC)	8,100	0%	**7.41%**	20–28 wks
HBM3e (SK/Samsung/Micron)	4,950	0%	**6.67%**	16–24 wks
400 kV Substations (India)	2,200	75%	**56.62%**	36–48 mos
Municipal Water Rights	N/A	100%	11,250 ML	Seasonal

D. Finding 3: Tropical Cooling Penalty and Water Crisis

Table IV presents the cooling trade-off matrix. Under conventional evaporative air cooling at summer peak (>42°C), PUE reaches 1.65 and water consumption reaches **11,651 ML/yr** for 500k GPUs—equivalent to the annual water supply of ≈280,000 Indian households. This creates direct conflict with municipal water rights, as manifested during Chennai's May 2024 drought in Ambattur and Siruseri where data centres faced 40% municipal water cuts, forcing private tanker procurement and emergency SWRO (+16% opex). Furthermore, tropical ambient heat accelerates hardware failure modes: Coffin-Manson thermal fatigue in 2.5D interposer micro-bumps (2.8×), TDDB in HBM3e TSVs (3.4× at > 85°C), and Arrhenius laser-diode decay in 800G transceivers (3.8×) [27]. Transitioning to Direct-to-Chip (D2C) liquid cooling slashes water consumption by **94.6%** (11,651→631 ML/yr) and peak power by **172 MW** (PUE 1.18).

TABLE IV
COOLING TECHNOLOGY TRADE-OFF MATRIX UNDER TROPICAL CLIMATES (500K GPUS) [SOURCES: [15], [19], [27]]

Cooling System	Ann. PUE	Peak PUE	WUE	Water (ML)
Traditional Air (Evap.)	1.45	1.65	1.80	11,651.2
Rear-Door Heat Ex.	1.35	1.45	1.00	6,031.5
Direct-to-Chip (D2C)	**1.18**	**1.25**	**0.12**	**630.9**
Single-Phase Immersion	1.12	1.18	0.03	150.8

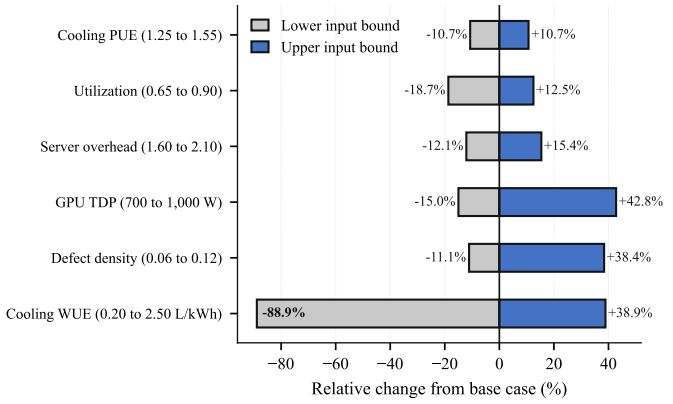


Fig. 3. One-at-a-time sensitivity tornado: cooling efficiency and host power overhead dominate peak grid demand across 500k GPU scenarios.

E. Finding 4: Cross-National Compute Gap and Fiscal Transfer

India's sovereign GPU fleet of 34,000–45,000 accelerators represents ≈0.3% of the US fleet (4M units, ≈1,200 ExaFLOPS) and <1% of China's 1.5 million units [28]. The 500k baseline elevates India to ≈245 ExaFLOPS—closing the gap with China to within 2×—but requires managing deep structural dependencies under iCET [29]. Furthermore, IndiaAI compute subsidies at Rs. 65 per GPU-hour (≈$0.78/hr) against commercial rates of $2.00–2.80/hr [16] represent a 72% fiscal transfer sustainable only while underlying supply chains remain open.

VI. STRATEGIC ROADMAP AND POLICY IMPLICATIONS

DCM results motivate a three-tier intervention architecture ranked by feasibility horizon and sovereignty leverage.

Tier 1 – Build Domestically (2026–2028): (i) *Geographic Diversification*: mandate 15–20% capital rebates for greenfield AI clusters outside Mumbai, targeting Mumbai below 35% of national capacity by 2028; pre-equip five AI Power Corridors

TABLE V
INVESTMENT ARCHITECTURE: THREE TIERS, FIVE PILLARS (2026–2035) [SOURCES: [6], [7], [14], [22], [30]]

Intervention Area	Public Rs. Cr	PPP Rs. Cr	Lead Agency
AIZs & AICA (P3/P5)	2,500	5,000	DPIIT, MeitY
Power Corridors + BESS (P2)	15,000	65,000	MoP, SECI
GPU Cluster Expansion (P1)	20,000	80,000	MeitY, IndiaAI
Packaging JVs/Gallium (P1)	10,000	40,000	ISM, CSIR
SMR/Nuclear Baseload (P2)	25,000	50,000	DAE, NPCIL
Talent & R&D (P5)	15,000	5,000	MoE, DST
Connectivity/Security (P4)	8,000	12,000	DoT, NCIIPC
Total	**95,500**	**2,57,000**	

with 400/220 kV infrastructure rated for 5 GW aggregate load. (ii) *D2C Cooling Standard*: condition IndiaAI subsidies on PUE $\leq$ 1.25 and closed-loop dry heat rejection, saving 11 billion litres/yr at 500k scale. (iii) *Fast-Track Substations*: create a 9-month CEA approval window for $\geq$50 MW compute substations; establish an AI Infrastructure Clearance Authority (AICA) under PMO for 90-day composite clearances. (iv) *AI Infrastructure Zones (AIZs)*: designate 20 pre-cleared zones across 10 states with pre-built 400 kV substations and 100 Gbps dark fibre.

Tier 2 – Develop Through Partnerships (2028–2031): (i) *Advanced Packaging JVs*: expand Tata Electronics (Morigaon) and Micron Sanand ATMP toward 2.5D silicon interposer via JVs with ASE Group and Ibiden under Semicon India 2.0, reducing India's 7.41% CoWoS dependency below 1% by 2032. (ii) *Critical Mineral Refining*: scale CSIR-IMMT hydrometallurgical gallium extraction from NALCO/Hindalco bauxite red mud to 50 Mt/yr of 6N-purity gallium [30]. (iii) *Sovereign Chiplet Consortium*: mobilise 120,000 VLSI engineers into a UCIe-standard RISC-V chiplet consortium fabricated at Tata-PSMC Dholera. (iv) *Captive Renewable Baseload*: transition AI campuses to Round-the-Clock hybrid solar-wind-BESS power, supplemented by SMRs under the SHANTI Act targeting first nuclear delivery by 2030.

Tier 3 – Manage External Dependencies (2031–2035): (i) *Foundry Reservations*: secure multi-year sovereign wafer allocations at TSMC/Samsung, commercially manageable at 1.60% of TSMC output. (ii) *HBM Supply Treaties*: advance-purchase reservations for HBM4 under the QUAD Semiconductor Supply Chain Initiative. (iii) *EAR Contingency Playbook*: formal technology-shock response plans covering various US export restriction scenarios.

A. Investment Architecture

Table V summarises capital requirements. The public-sector outlay of ≈Rs. 95,500 crore confirms the "N-2 asymmetry" vs. the US Stargate ($500 billion), but India's 1.4-billion-user market and neutral geopolitical position can attract Rs. 2,57,000 crore in private and multilateral capital.

B. Measurable Indicators and Key Performance Metrics

To operationalise IAIRF 2.0 as an active monitoring instrument, each pillar requires trackable KPIs: **P1 (Compute):** Sovereign GPU fleet size; domestic ATMP throughput; VLSI IP registrations; startup compute subsidy uptake. **P2 (Energy):** Grid-connection lead time for $\geq$50 MW compute; captive RTC renewable share; commissioned BESS capacity; average power tariff. **P3 (Physical):** Mumbai capacity share (<35% target); D2C liquid-cooled share; PUE; operational AIZs. **P4 (Connectivity):** Inter-metro backbone Tbps; DPDPA implementing rules; OT cybersecurity compliance [22]; submarine cable landings. **P5 (Governance):** AICA 90-day clearance rate; IndiaAI fund disbursement %; annual AI PhD completions.

C. GDP Impact and Economic Multiplier Analysis

NITI Aayog projects AI will contribute $500 billion to India's GDP by 2030. Stanford HAI [28] demonstrates that each dollar of sovereign compute generates $3–5 in downstream economic value within five years. Deploying 500k GPUs (245 ExaFLOPS) enables training 50+ frontier foundation models annually, processing 2 billion daily public queries, and supporting 3,500+ startups—yielding $120 billion in direct AI GDP contribution by 2030 and compounding to $2 trillion by 2047.

VII. LIMITATIONS AND FUTURE RESEARCH

The Demand-Cascade Model rests on three core assumptions that bound generalisability. First, static annual cluster utilisation ($U = 0.80$) omits dynamic diurnal inference surges where power draws spike 20–40%, potentially inducing localized grid stress. Second, TSMC's proprietary defect density parameters for its custom 4N process remain confidential; the Murphy model utilizes public academic baselines ($D_0 = 0.08$ cm^{-2}), introducing an estimated ±8–12% uncertainty envelope around net wafer starts. Third, the geopolitical trajectory of US Export Administration Regulations is exogenous; continued H100/B200 hardware access under iCET remains subject to evolving bilateral policy shifts. Furthermore, capital expenditure baselines assume static 2026 equipment pricing without discounting for manufacturing learning curves. Future research will couple DCM telemetry directly to dynamic AC Optimal Power Flow (ACOPF) bus-level simulators across regional transmission rings, incorporate diurnal workload traces from the IndiaAI Mission's monitoring portal, and generalize the five-pillar framework to ASEAN digital middle powers.

VIII. CONCLUSION

This paper has presented IAIRF 2.0 and the Demand-Cascade Model as complementary instruments for evidence-based AI infrastructure planning in India. The empirical results challenge prevailing assumptions: silicon wafer fabrication is not India's binding constraint; advanced packaging and metropolitan grid transmission are the true structural chokepoints; tropical cooling physics impose a structural water-security threat; and Mumbai concentration creates systemic fragility that software sovereignty cannot offset. Capital should flow to transmission corridors, CoWoS packaging JVs, D2C cooling mandates, and pre-cleared AIZs—not premature leading-edge foundries.

India possesses structural competitive advantages: unmatched engineering talent depth, a 1.4-billion-user multilingual data asset, Rs. 10,371 crore of committed public compute capital, and a geopolitical position coveted by both the United States and the Global South. The IAIRF 2.0 framework and three-tier roadmap provide a technically grounded path for converting these latent advantages into operational infrastructure before the window of competitive opportunity narrows. The AI race is ultimately an infrastructure race—won in the unglamorous work of clearing land, stringing transmission lines, negotiating packaging capacity, and dissolving bureaucratic red tape. India can win it—at the speed of transmission towers.

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